

Exploring Particle Dynamics through Simulation and Experimentation: A Portfolio

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This portfolio presents a series of exercises aimed at exploring various physical concepts, including projectile motion, restitution coefficients, and particle dynamics. The work is split into three distinct parts, each linked whilst focusing on separate aspects such as experimental and theoretical analysis using two programming languages: Python and MATLAB.

The first portfolio discusses numerical methods and uses the Euler method, which was compared with the Velocity-Verlet method as well, to simulate projectile motion across different environments and different conditions. These methods modelled the effects of gravity and air resistance whilst also taking into account initial conditions, such as initial velocity and angle of projection, etc. The work then grew to simulate the behaviour of a ball bouncing off a solid surface, adopting the “hard-sphere” model rather than the “soft-sphere” model [1] to simulate elastic collisions; while less accurate, this method was used due to its ease of implementation. Overall, the case study became a profound insight into the impact of timestep selection and how that impacts simulation accuracy. Also, I researched terminal velocities for various materials in various media, which provided context for the next batch of experiments.

The second portfolio transitions from theoretical simulations to experimental work, in which I designed an experiment to determine the restitution coefficient of a ball with respect to a surface, including the theory to describe these concepts. I dropped both a table-tennis and a tennis ball on two types of surfaces, a domestic carpet, and a chunk of ‘Medium-Density Fiberboard’, and analysed their motion using particle tracking velocimetry (PTV), calculating the restitution coefficient by measuring their velocities before and after the bounce. The most important aspects of this were the definition of a solid experimental setup including lighting, filming angle and surface materials. This experiment also proved a good point to apply image processing to the dataset and to be able to generalize the code to use different kind of datasets thus evaluating whether the end results are reliable and accurate. These portfolios provide a holistic study of the effect of moving particles, through numerical modelling, experimental work and literature review (below) to establish an encompassing idea around this topic. From the simulation of Portfolio 1, to the experimental setup of Portfolio 2, it shows the balance of methods used to complement each other to study how particles behave.

The Discrete Element Method/Modelling (DEM) has been used widely over the past three decades; Discrete Element Method/Modelling refers to a family of numerical methods for computing the interactions of a large number of particles such as granular materials between themselves and also various surfaces [2]. These simulations assist in the designing of machinery for industry and also for research. However many limitations exist by virtue of reducing computational power; hence, particle shapes are typically limited to less complex geometries such as circles and ellipses. The limitation also extend to physical forces such as friction which raise the question of validity to the models [3]. There are three major types of DEM methods used in literature which have fundamental differences [2]: contact dynamics (NC)/nonsmooth contact dynamics (NCS) [2, 4, 5, 6] where small elastic response times and displacements are neglected [6]; molecular dynamics (MD) [1, 7, 8] the particles are treated as rigid bodies, but the contacts between particles are assumed to be compliant and obeying a viscoelastic behaviour [2, 4, 5, 6]; granular element method (GEM) [2, 9, 10] overcomes one of the limitation previously discussed in the computational discrete mechanics of granular materials by allowing discrete elements to take realistic and complex granular shapes encountered in engineering and science (e.g., sand grains) [9].

DEM’s were initially intended as a tool for rock mechanics by Cundall [1, 2, 11] and so based on the soft sphere model Cundall and Strack [1] proposed, particles were seen to have two types of motion: translational and rotational, which were thereby determined by Newton’s Second Law of Motion [2], similar to the equations one used for the governing of particle motion shown in Portfolio 1 below. The creation of the shape of particles has a significant impact on the interactions between particles in the context they need to be used in, and so many models can be found in literature. The choice between 2 dimensional [12] and 3 dimensional shapes may also be of significance depending on the context they are to be employed.

Portfolio 1 and 2 aim to use modelling techniques based in theory and literature and refined using experimentation.

Portfolio 1

1. All code can be found in `\portfolio_work_1 (1).py`", all code is in the same order as the analysis below.

2.

(a)

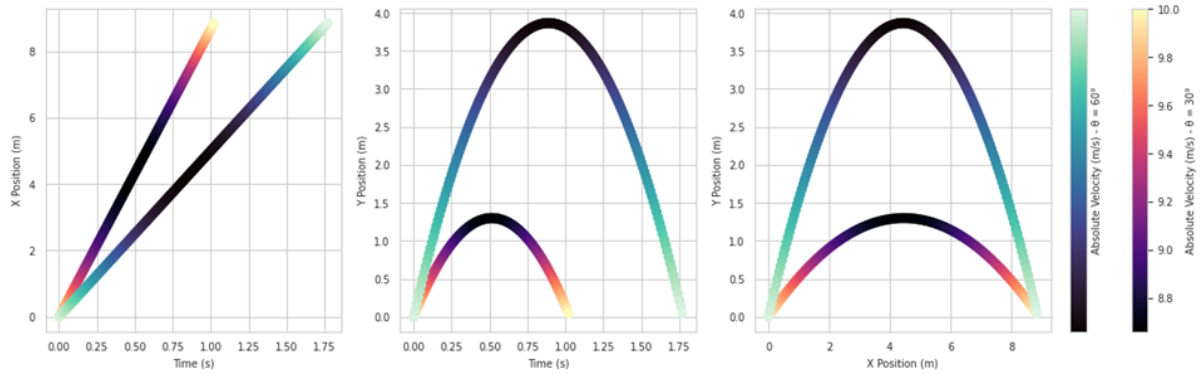


Figure 1: These graphs show the projectile motion of a particle launched from ground level, with an angle of projection of 30° , an initial velocity of 10 m/s and varying angles of projection (θ) with no drag effects.

(b) The results shown in Fig. 1 are expected. When taking the problem at first glance it may seem counter-intuitive as to how a projectile thrown at double the angle to the horizontal can travel the same distance; however, when solving the problem analytically, it is confirmed that both travel the same distance (8.83 m); this is due to if it was projected more vertically, it has little relative horizontal velocity but spends much longer in the air and if projected at more horizontally, the opposite is true; and so the projectile travels the same distance (displacement is the correct term).

(c)

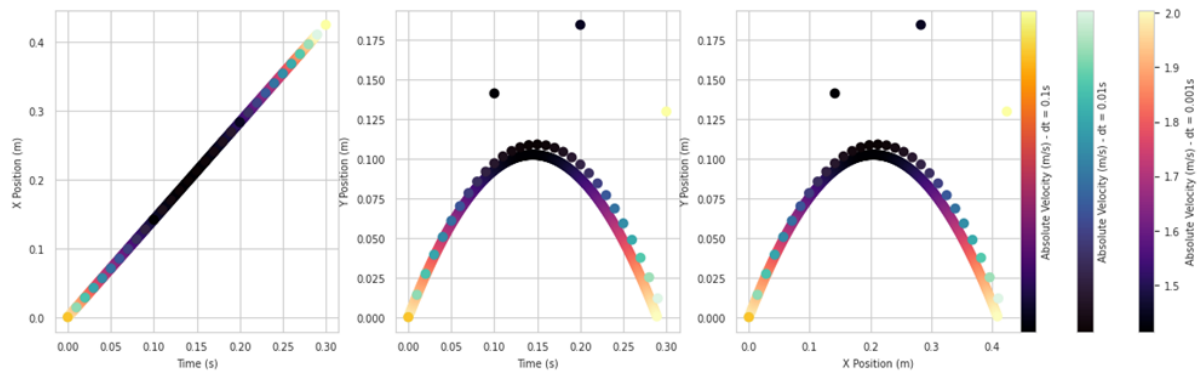


Figure 2: These graphs show the projectile motion of a particle launched from ground level, with an angle of projection of 45° , an initial velocity of 2 m/s and varying time steps (dt) with no drag effects

Fig. 2 clearly highlights the effects of different time steps (dt) when using the Euler method. Rather intuitively, it is evident that the smaller the ' dt ', the greater the accuracy of the model, however this comes with the drawback of longer a computational time. The Euler method also has a first-order accuracy which states that the error in the method is proportional to the size of the time step and so when ' dt ' was increased by an order of magnitude, so did the error. The magnitude of the discrepancies for the vertical position of the particle are extremely large when quantitatively compared; the maximum height reached by the particle using the ' dt ' value of 0.1 s was at a different time than to the other two time steps and was also 1.5 times greater than each of the maximum heights at time steps of 0.001 and 0.01 s.

(d)

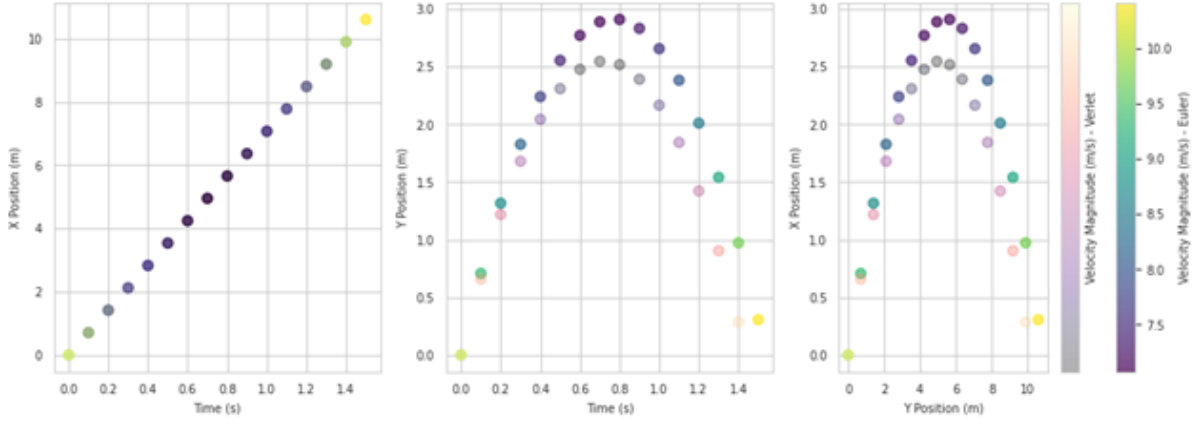


Figure 3: These graphs show the projectile motion using the Euler and Velocity-Verlet method of a particle launched from ground level, with an angle of projection of 45° , an initial velocity of 10 m/s and time steps of 0.1 s

Both the Verlet and Euler method are widely used, the latter more frequently; this is due to the Verlet's method having a higher level of accuracy; seen especially when larger time steps are used (Fig. 3). The greater accuracy in the Verlet method comes as a result of it using values for position and velocities for both previous and current time steps as opposed to only using said values from the current time step. The Verlet method also has second-order accuracy and so the error in the numerical solution decreases proportionally to the square of the time step. And so decreasing the time step by an order of magnitude, decreases the error by two orders of magnitude in contrast to just one (as the Euler method does). This accuracy can be seen when comparing the maximum heights reached by both methods to that of the analytical answer. The Euler method provided a maximum height of 2.91 m; the Verlet method, a value of 2.55 m; and the analytical solution, a value of 2.55 m.

3.

(a)

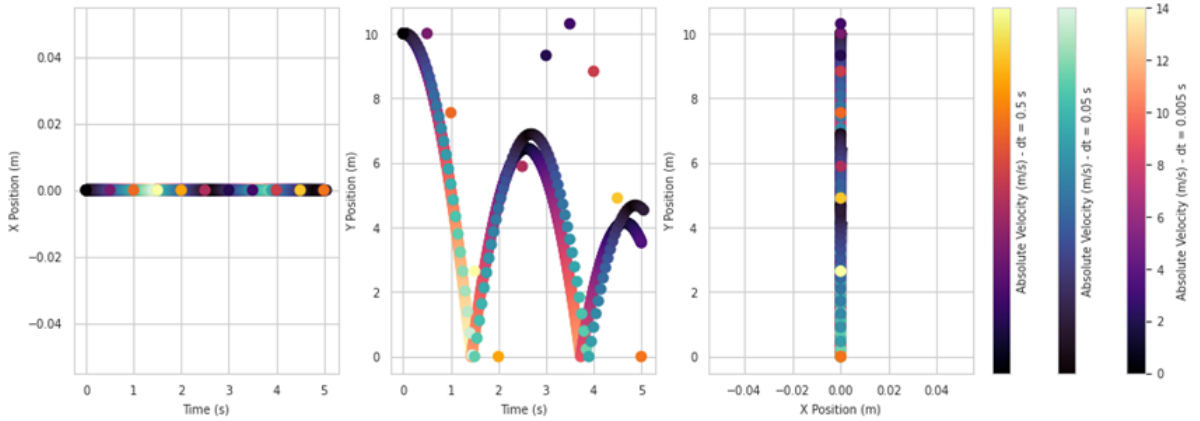


Figure 4: These graphs show the projectile motion using the Euler method for a steel ball being dropped from 10 m above ground level with the effects of bouncing, no drag effects, and varying time steps (dt)

As explained previously, the Euler method has first-order accuracy and so the error decreases linearly with the time step chosen; this is highlighted in both Fig. 4 and Fig. 2, however it is most evident when observing the maximum heights of the particle in Fig. 4. An error occurs whereby when too large a time step is used, the particle is seen to gain energy and reach heights above that which it was dropped. When a time step of 0.5 s was used, the particle achieved a maximum height of 10.3 m (and so 0.3 m higher than its starting position) in contrast to the other time steps; and so an error is exposed even when a coefficient of restitution of less than 1 (0.8) is used. B.

(b) Discussed earlier, the problem of finding an appropriate 'dt' value, and in most cases, is very subjective to the accuracy that is trying to be achieved and more so, the length of time that the simulation is. In the example carried out, both 'dt' values of 0.05 s and 0.005 s were appropriate (for the simulation time chosen) and gave very similar results and so with the intent of faster computation, it is recommended a 'dt' value of two orders of magnitude less than the simulation time should be used.

(c)

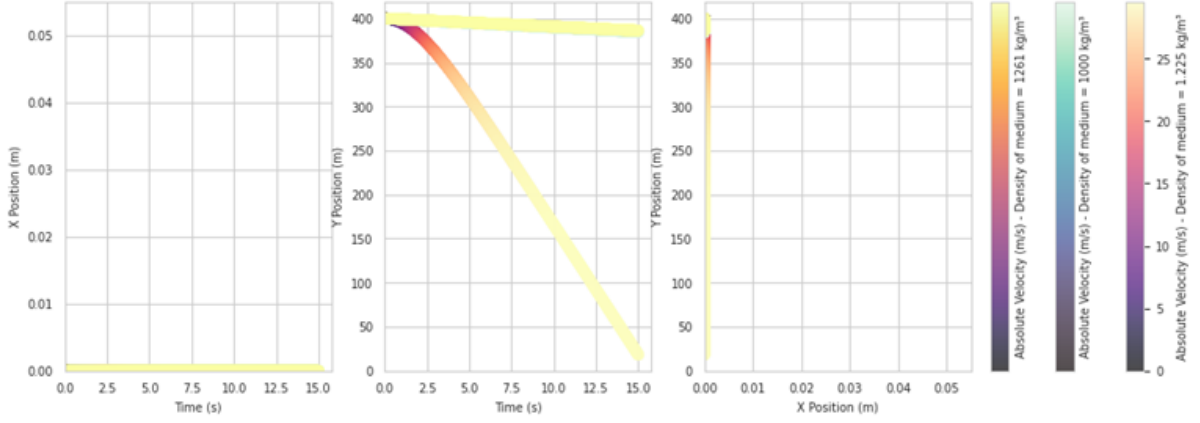


Figure 5: These graphs show the projectile motion using the Euler method for a steel ball (of a 5 mm diameter) being dropped from 10 m above ground level with the effects of bouncing drag effects, and varying densities of medium

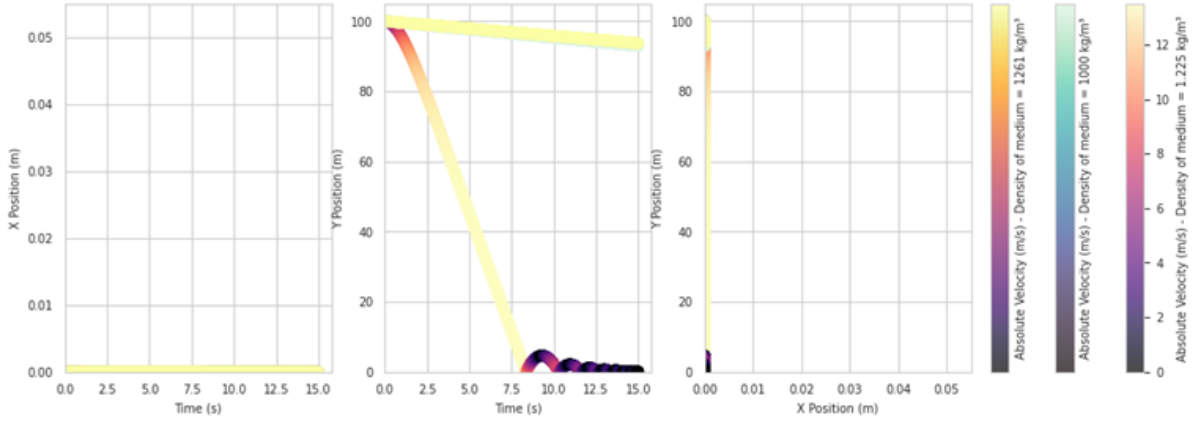


Figure 6: These graphs show the projectile motion using the Euler method for a nylon ball (of a 5 mm diameter) being dropped from 10 m above ground level with the effects of bouncing drag effects, and varying densities of medium

The mass used for the steel and nylon balls were found using their respective densities (7.7×10^3 and $1.6 \times 10^3 \text{ kgm}^{-3}$) and the volume of the balls (from the 5 mm diameter specified). Densities of air, water and glycerine were also used (which were 1.225 , 1000 and 1261 kgm^{-3} respectively) and a typical drag coefficient for a sphere was used (0.47). The drag force was then calculated and implemented into the calculations for the x and y velocities. The model successfully produced all the terminal velocities (bar one), when compared to their analytical counterparts, in the different media for both material balls; the values obtained were:

Medium	Steel Ball - Terminal Velocity (m/s)	Nylon Ball - Terminal Velocity (m/s)
Air	29.57	13.50
Water	1.04	0.47
Glycerine	0.92	0.42

Figure 7: Terminal velocities (m/s) of a 5 mm steel and nylon ball in three different media

The value calculated for the steel ball in air was 29.57 m/s (Fig. 7) when it should be 30.61 m/s . This error is a result of the condition used to calculate the terminal velocities in the simulation; the particle is said to have reached terminal velocity when the difference between the new and current velocity is less than 0.0001 s – this can be changed and is encouraged within comments of the code. Making the condition more stringent may also need to be accompanied with an increased simulation time and dropping height to allow for terminal velocity to now be reached (as seen in Fig. 5 and 6). The analytical equation used to compare the simulated terminal velocities was: $v = \sqrt{\frac{2mg}{AC_D\rho}}$

Portfolio 2

A home experiment, using a full factorial design, was designed and conducted to test the effect two different variables had on the coefficient of restitution (COR). These variables were: the material of the ball; and the surface of the floor the ball was being dropped on.

A black background was selected along with dark coloured flooring, in order for the greyscaling and binarization to be effective. The two floor materials were a piece of MDF (Medium-Density Fiberboard) and a domestic carpet. This allowed for the two extremes of floor material to be experimented on: a hard, and soft floor, respectively.

The ball types differed also; a tennis ball and table tennis ball were utilised; this also allowed for the two extremes of ball types to be measured: soft and hard, respectively. Both were hollow to eliminate differences based on the density of the balls; they were also bright in colour to create a strong contrast between themselves and the surrounding environment; making the binarization and greyscaling further effective.

The filming angle chosen was to be roughly half way between the drop heights of the balls and the surface they were falling on which created a good understanding of the height they were dropped at; and so the starting position and the ending position can be seen in the videos without the need to change camera positions.

The coefficient of restitution for each run was found by taking the mean average for every individual coefficient of restitution calculation that was done for each two consecutive bounces.

The calculation for each individual coefficient was done by using this equation: *Coefficient of restituion* = $\sqrt{\frac{h_{n+1}}{h_n}}$, where h refers to the initial height, and then height of each of subsequent peak.

Results

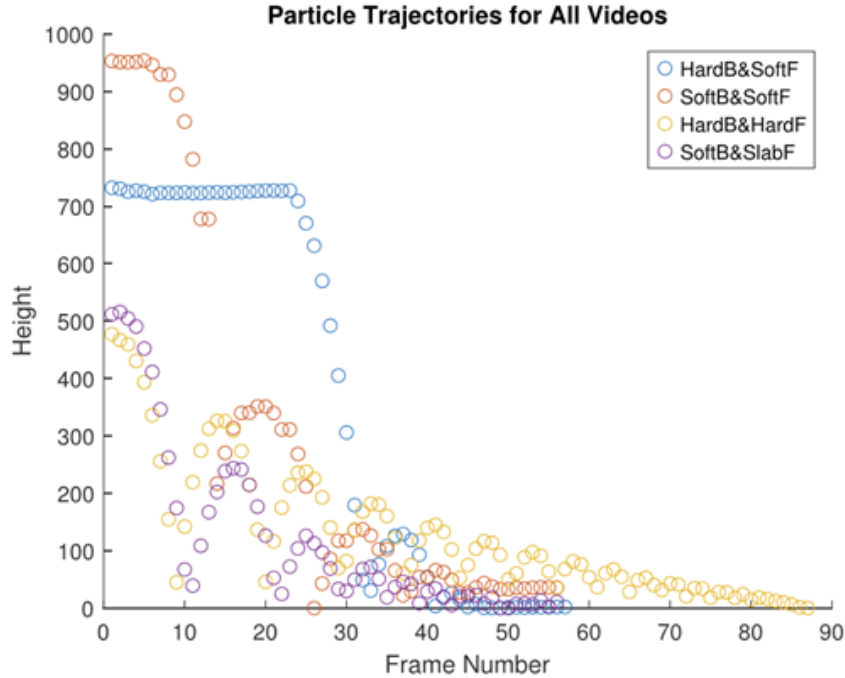


Figure 8: displays the movements in the y direction of the particles. 'Hard and Soft B' refer to the Tennis and Table-tennis ball respectively. 'Hard and Soft F' refer to the MDF and carpet, respectively.

	Steel Ball - Tennis Ball (Soft B)	Table-tennis Ball (Hard B)
Carpet (Soft F)	0.62587	0.65741
MDF (Hard F)	0.73763	0.91096

Figure 9: displays restitution coefficients for changing variables

Using the Fig. 8 above and the coefficients of restitution that the MATLAB code presents in Fig. 9, it is clear to see that restitution coefficients rely significantly on the surface of the floor that the particle is dropping on; the values also indicate the harder, MDF, surface provides more elastic collisions in comparison to the softer carpet. On both surfaces, the hard ball (table tennis ball) has a higher COR than the soft ball (tennis ball). This suggests that the hard ball retains more of its kinetic energy after the bounce, resulting in a higher rebound as opposed to the soft ball which tends to deform more upon impact, reducing its ability to bounce back as efficiently.



Figure 10: displays an example of a frame of the binarized and greyscaled video

There may be inaccuracies in the presented results, influenced heavily by limitations in the experimental setup and measurement techniques; this is clearly seen in Fig. 10 where many points of tracking is occurring. The points at the top of the figure display the fingers of the hand that released the ball and the points of interest at the bottom of the figure are lighter points of the flooring; this may skew the results observed. Another point of error may be where the ball bounces consecutively very fast as it nears its point of being stationary and so the tracking system may find it difficult to differentiate where the ball is in the specific frame.

As seen in Fig. 8 and Fig. 9, the coefficient of restitution can defer significantly between the same ‘type’ of object (a ball) where the values cover around 40% of the range that the COR can take; and so common CORs used in literature, such as 0.8, may be incorrect in terms of which environments they are being used in; however, the difference in COR value for each variable may be less significant when observing the scope of its effect in an equation using its value. And so considerations of a trade off can be seen where, the effort of obtaining a COR that is specific to one’s system may not offset the inaccuracy of using a generic COR from literature.

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